

Experienced Senior Software Engineer and Engineering Manager with over 10 years of expertise in developing and optimizing scalable applications and data systems. Proficient in Python, Java, Golang, and cloud technologies (AWS, Azure, GCP), with strong hands-on experience in building RESTful APIs, designing microservices, and engineering ETL pipelines for high-performance data processing. Proven track record in leading cross-functional teams, architecting cloud-based solutions, and driving Agile development cycles.

## Work History

### Senior Software Engineer / Engineering Manager

Crown Consulting, Inc, Pine Brook, NJ

February 2021 - February 2025

- Developed robust and scalable Python applications for **data processing**, automation, and API integration, leveraging Python libraries such as Pandas, NumPy, and requests to streamline workflows and ensure high efficiency.
- Built and maintained **RESTful APIs** with Python (Flask, FastAPI) integrated with AWS Lambda, ensuring high availability and low latency in serving requests while implementing strong authentication and authorization practices.
- Developed Python-based **web applications** using Flask and Django, focusing on scalability, performance, and robust backend architecture.
- Integrated Python applications with **Snowflake** by utilizing the Snowflake Connector for Python to enable seamless data loading, transformation, and retrieval, optimizing ETL pipelines.
- Developed and optimized complex SQL queries within Snowflake, using Python's **snowflake-connector-python** to automate data extraction and reporting, reducing manual effort and improving query performance.
- Utilized **Django Ninja** to implement complex data validation and parsing logic for incoming API requests, ensuring data integrity and reducing errors in application workflows.
- Developed **Power BI** reports and dashboards by integrating Python scripts to fetch and preprocess data from various sources, ensuring accurate and real-time insights.
- Automated **data extraction** and **transformation** for Power BI reports using Python, leveraging libraries like pandas and pyodbc to integrate with SQL databases and APIs.
- Designed and deployed **RESTful APIs** using Django Ninja for an **e-commerce platform**, enabling smooth integration with third-party systems and enhancing user experience with high-performance endpoints.
- Designed and implemented high-performance RESTful APIs and **microservices** using **Golang**, enhancing system scalability and reliability, and enabling seamless integration with frontend and third-party services.
- Developed concurrent **Golang applications** leveraging goroutines and channels for efficient data processing, improving application performance and responsiveness under heavy loads.
- Developed and maintained full-stack web applications using **React**, **Node.js**, and Python, ensuring efficient communication between front-end and back-end components while optimizing for scalability and performance.
- Designed and implemented RESTful APIs with **Node.js** and Python, enabling seamless

integration with third- party services and ensuring high availability and reliability for production environments.

- Engineered and optimized **NoSQL** databases (such as MongoDB, Cassandra, or Redis) to support high- availability, low-latency access patterns for Python applications.
- Developed and optimized **PostgreSQL** database schemas for high-performance applications, including complex data models, indexing strategies, and partitioning to enhance query speed and system scalability.
- Designed and implemented complex **SQL** queries for large-scale databases, optimizing performance through indexing, query optimization, and advanced join techniques, enabling fast and accurate data retrieval in high- volume environments.
- Integrated **JavaScript** and **TypeScript** with Python-based back-end services to enhance web functionality, ensuring seamless communication between the front-end and back-end systems.
- Developed and maintained **RESTful APIs** using Spring Boot, enabling efficient data flow between Python services and front-end applications built with JavaScript and TypeScript.
- Facilitated **Scrum** ceremonies, including sprint planning, retrospectives, and daily stand-ups, to ensure efficient collaboration across development teams and delivered Python-based features on time and within the sprint scope.
- Worked closely with the Product Owner and **Scrum Master** to define and prioritize Python-based tasks in the product backlog, ensuring alignment with business goals and timely delivery of incremental features.
- Developed and optimized **ELT workflows** with Python, ensuring data transformation occurred within the data warehouse, reducing data movement costs and improving processing efficiency.
- Led **Agile development** cycles, managing sprint planning, daily stand-ups, and cross-functional collaboration to maintain project momentum, delivering new features on time and within scope.
- Managed version control and code collaboration on **GitHub** by overseeing repositories, branching strategies, and pull request workflows to ensure seamless code integration and release management.
- Architected **cloud-based solutions** using AWS services (Lambda, S3, EC2, RDS, DynamoDB) to ensure highly available, cost-effective, and scalable systems, enabling seamless integration with Python applications.
- Optimized data pipelines using Python and **AWS** services, automating data extraction, transformation, and loading (**ETL**) processes, which reduced data processing time by 30%.
- Collaborated with **DevOps** teams to containerize Python applications using **Docker**, enabling easier deployment and scalability in cloud environments.
- Implemented continuous integration and continuous deployment (**CI/CD**) pipelines using AWS CodePipeline, CodeBuild, and GitHub Actions to automate testing and deployment of Python applications in a cloud environment.
- Utilized Python's **Pandas**, **NumPy**, and **Scikit-learn** libraries to build **machine learning models** for predictive analytics in telecom network traffic analysis.
- Contributed to the development of a **real-time data processing** pipeline, utilizing Python's **asyncio** and **Celery** for task scheduling and managing concurrent tasks efficiently.
- Designed and implemented scalable **BI/DW architecture**, utilizing data warehousing principles to streamline data integration, storage, and reporting processes for large datasets across multiple sources.
- Led end-to-end **ETL pipeline** development using Python, integrating data from diverse

sources (SQL, NoSQL, APIs), transforming it into structured formats, and loading it into centralized data repositories for business analytics.

- Designed flexible, scalable schema for **NoSQL** systems, enabling rapid iterations and seamless handling of unstructured data.
- Led **database schema design** and optimization in PostgreSQL, ensuring efficient queries, indexing, and data integrity within Django's ORM framework.
- Collaborated with cross-functional teams to design and implement scalable **database solutions**, incorporating Python-based ORM (SQLAlchemy) and advanced SQL design principles to improve data access patterns and minimize latency.
- Implemented **microservices** in Go, focusing on optimized memory usage, efficient concurrent processing using Goroutines, and robust error handling to ensure system reliability.
- Collaborated with backend developers to integrate **RESTful APIs** and ensure seamless data flow between the front-end and back-end services.

## Senior Software Engineer

BAE Systems, Inc, Totowa, NJ

August 2016 - November 2020

- Implemented advanced **ETL** processes to load, transform, and analyze large datasets from various sources into Redshift, ensuring data integrity and availability for business analytics.
- Automated data workflows with **Python scripts** to schedule, monitor, and manage data pipelines on Snowflake, ensuring data accuracy and reliability for real-time analytics.
- Engineered data ingestion pipelines using AWS **Kinesis** to handle large-scale streaming data, ensuring real-time processing and near-instant analytics on data streams.
- Developed efficient **SQL** queries and database schemas to support complex reporting and transactional workloads in Python-based systems.
- Automated complex data workflows and data transformation tasks using **Python** scripts, significantly reducing manual intervention and enhancing process efficiency.
- Utilized **Spring Boot** to build microservices that were integrated with Python-based systems, optimizing scalability and fault tolerance across the infrastructure.
- Applied Agile principles within a Python development team, continuously improving workflow processes using **Scrum** and **Kanban** techniques to achieve faster release cycles and higher-quality software.
- Collaborated with **Scrum teams** to define and implement Python unit tests and CI/CD pipelines, ensuring code quality and smooth integration with ongoing development tasks and releases.
- Developed robust, scalable web applications using **Django**, adhering to best practices such as DRY (Don't Repeat Yourself) and modular architecture for maintainability.
- Designed and optimized **RESTful APIs** using Django REST Framework (**DRF**), ensuring high-performance endpoints with advanced query handling and serialization.
- Built scalable **RESTful APIs** using **Node.js** and **Express**, handling thousands of concurrent requests while maintaining high availability and fault tolerance.
- Utilized asynchronous programming techniques in **Python**, **Spring Boot**, and **JavaScript** to optimize resource handling and reduce latency in complex data processing systems.
- Designed and developed responsive, user-centric front-end applications using **Vue.js**, improving the user experience and application performance by integrating efficient state management (**Vuex**) and component-based architecture.
- Designed and maintained containerized microservices using **Docker** and orchestrated them

with **Kubernetes**

- to ensure seamless scaling, self-healing, and high availability.
- Designed and developed **Python**-based solutions for telecom data analytics, processing large volumes of network performance metrics to identify trends and optimize resource allocation.
- Facilitated backlog grooming sessions and sprint reviews, ensuring that project requirements were clearly defined and prioritized based on stakeholder needs.

## Software Engineer

PGIM, Newark, NJ

March 2014 - August 2016

- Designed and deployed **cloud-native** applications using Microsoft Azure services, including Azure App Services, Azure Functions, and Azure SQL Database for scalable, serverless architectures.
- Architected and deployed scalable applications using **GCP** services like Compute Engine, Cloud Storage, and Cloud Pub/Sub, ensuring optimal performance and high availability.
- Designed and implemented scalable **microservices** using Scala, ensuring efficient and maintainable code for high-performance applications.
- Leveraged **Java-based** backend systems to build robust, cross-platform services, integrating seamlessly with Python applications for full-stack development.
- Developed Java libraries for complex data processing tasks, interfacing with Python via **RESTful APIs** for smooth data exchange.
- Implemented **authentication** and authorization mechanisms using JWT and OAuth2 for secure user sessions and data privacy.
- Automated deployment processes on **Linux** environments using shell scripting, significantly improving system efficiency and reducing manual intervention.
- Streamlined development pipelines on **Linux** servers by leveraging shell commands to automate testing, builds, and deployments in Python, resulting in a 20% reduction in deployment time.

## Education

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Bachelor of Science in Computer Science July 2013

Stockton University, Absecon, NJ